VERITAS UNIVERSITY, ABUJA

DEPARTMENT OF SOFTWARE ENGINEERING

VUA-SEN307 DATABASE SYSTEMS

WEEK 4 PRACTICAL — LAB 2: DATABASE DESIGN AND NORMALIZATION

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Platform	Windows laptop

Purpose

This write-up follows the Week 4 practical handout. It identifies the course-registration entities and relationships, demonstrates normalization from 1NF to 3NF, and presents a final relational schema. The ER diagram reference and SQL setup script are supplied as separate files in the submission pack.

1. Entities, Attributes and Primary Keys

Entity	Suggested attributes	Primary key
Department	dept_id, dept_name	dept_id
Student	student_id, matric_no, full_name, dept_id	student_id
Lecturer	lecturer_id, lecturer_name, dept_id	lecturer_id
Course	course_id, course_code, course_title, dept_id	course_id
Enrolment	student_id, course_id, session	(student_id, course_id, session)
TeachingAssignment	lecturer_id, course_id, session	(lecturer_id, course_id, session)

TeachingAssignment is included to represent the many-to-many relationship between lecturers and courses and to record the session in which a lecturer teaches a course.

2. Relationships and Cardinalities

Relationship	Cardinality	Implementation
Department — Student	One-to-many (1:M)	Student.dept_id is a foreign key.

Relationship	Cardinality	Implementation
Department — Lecturer	One-to-many (1:M)	Lecturer.dept_id is a foreign key.
Department — Course	One-to-many (1:M)	Course.dept_id is a foreign key.
Student — Course	Many-to-many (M:N)	Resolved through Enrolment.
Lecturer — Course	Many-to-many (M:N)	Resolved through TeachingAssignment; session is stored here.

3. Unnormalized Registration Sheet (UNF)

The original sheet has multi-valued cells for course codes, course titles, lecturers and lecturer departments. A student can appear to register for several courses in one row, so the values are not atomic.

matric_no	student_name	student_dept	course_codes	course_titles	lecturers	lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems, Software Architecture	Dr A. Okafor, Dr M. Bello	Software Eng., Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database Systems, Operating Systems	Dr A. Okafor, Mrs F. Eze	Software Eng., Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

4. First Normal Form (1NF)

Fix: split every multi-valued cell so that each row contains one student–course pair and each cell contains one value. Use the composite key (matric_no, course_code, session), because a student can take multiple courses and the session distinguishes registrations over time.

matric_no	student_name	student_dept	course_code	course_title	lecturer	lecturer_dept	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

5. Second Normal Form (2NF)

Fix: remove partial dependencies from the composite key. Student details depend on matric_no, course details depend on course_code, and registration details depend on the complete student–course–session key.

Student_2NF (matric_no PK, student_name, student_dept)

matric_no	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Course_2NF (course_code PK, course_title)

course_code	course_title
SEN307	Database Systems
SEN301	Software Architecture
COS301	Operating Systems

Enrolment_2NF (matric_no FK, course_code FK, session; composite PK)

matric_no	course_code	session
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

6. Third Normal Form (3NF)

Fix: remove transitive dependencies by placing department details in Department and linking Student, Lecturer and Course to it using dept_id. Lecturer-course-session assignments are stored separately because the handout states that a course may have multiple lecturers across sessions.

Table	Columns and key details
Department	dept_id INT PK; dept_name VARCHAR(100) UNIQUE NOT NULL
Student	student_id INT PK; matric_no VARCHAR(30) UNIQUE NOT NULL; full_name VARCHAR(100) NOT NULL; dept_id INT FK → Department.dept_id
Lecturer	lecturer_id INT PK; lecturer_name VARCHAR(100) NOT NULL; dept_id INT FK → Department.dept_id
Course	course_id INT PK; course_code VARCHAR(20) UNIQUE NOT NULL; course_title VARCHAR(150) NOT NULL; dept_id INT FK → Department.dept_id
Enrolment	student_id INT FK → Student.student_id; course_id INT FK → Course.course_id; session VARCHAR(20); composite PK (student_id, course_id, session)

Table	Columns and key details
TeachingAssignment	lecturer_id INT FK → Lecturer.lecturer_id; course_id INT FK → Course.course_id; session VARCHAR(20); composite PK (lecturer_id, course_id, session)

Note: The handout does not provide unique IDs for departments, lecturers or courses. The IDs and TeachingAssignment junction table above are design choices to make the relational schema implementable and to represent the relationships described in the scenario.

7. Submission Checklist

• Normalization write-up: this PDF.

• SQL schema/setup: lab2_schema_and_mysql.sql.

• ER diagram reference: lab2_erd_reference.pdf (use it to recreate the diagram in MySQL Workbench and export the required lab2_erd.png).

• Workbench model: create and save lab2_model.mwb in MySQL Workbench; this native model must be saved by you in the application.

• Copy the files into the repository's week04 folder, then run git add ., git commit -m "Week 4: ER diagram and normalization to 3NF", and git push.